

Malcolm Nichols

Applied AI Engineer | LLM, NLP & Computer Vision Systems

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📄 Professional Summary

Applied AI Engineer with experience building and deploying production-grade AI systems using LLMs, NLP, OCR, computer vision, and scalable MLOps infrastructure. Proven track record designing and deploying enterprise AI systems, including Retrieval-Augmented Generation (RAG), clinical document intelligence, OCR/NER pipelines, predictive modeling, and computer vision automation. Experienced in deploying AI workloads in Databricks using MLflow, model serving endpoints, vector search, and cloud-native data platforms within regulated healthcare environments.

💡 Skills

AI & Generative AI

- NLP/NER
- OCR
- RAG
- LLMs, Embeddings
- Agentic AI
- Prompt Engineering
- Transformers
- Prompt Orchestration
- Inference Pipelines

Programming & Frameworks

- Python
- SQL
- REST APIs
- Spark
- Pytorch, TensorFlow
- Scikit-Learn
- Model Evaluation

Data & Cloud Platforms

- Databricks
- Vector Databases
- Distributed Processing
- Data Pipelines
- Azure
- Delta Tables

AI Infrastructure / MLOps

- MLflow
- CI/CD (Azure DevOps)
- Agile (Scrum)
- Model Serving Endpoints
- Git
- Docker
- Unity Catalog

🏢 Professional Experience

Univera Healthcare

AI Engineer

01/2024 – Present | Buffalo, NY

- Engineered scalable distributed data pipelines processing millions of documents, improving throughput from 20 to 10 minutes per 100K files (50% performance increase).
- Architected and deployed a production OCR/NER clinical document intelligence pipeline leveraging NLP, rule-based extraction, and document intelligence methods, reducing manual review time by 40% and improving compliance reporting accuracy.
- Designed and productionized an enterprise Retrieval-Augmented Generation (RAG) system leveraging vector search, embeddings, prompt orchestration, and LLM inference, reducing internal document lookup time by 55%.
- Developed predictive healthcare AI models for HEDIS compliance optimization, improving preventive outreach efficiency by 20%.
- Developed and deployed a YOLO-based computer vision automation system for QA workflows, reducing manual testing costs by ~\$50K annually through computer vision-based UI element detection.
- Built scalable MLOps and AI model deployment workflows using Databricks, MLflow, Unity Catalog, and real-time model serving endpoints.
- Presented AI strategy and research findings to executive leadership, influencing enterprise AI adoption.
- Led and mentored AI interns across production NLP and computer vision initiatives, conducting performance reviews and guiding enterprise AI project delivery.

Data Science Intern

06/2023 – 01/2024 | Buffalo, NY

- Designed and deployed the company's first Generative AI chatbot supporting HR and Workday documentation.
- Evaluated and implemented transformer architectures using PyTorch for enterprise NLP applications.
- Deployed APIs and production endpoints using Databricks model serving infrastructure.

🎓 Education

Master of Science, Artificial Intelligence, Maryville University of St. Louis

05/2024

Bachelor of Science, Computer Science, Maryville University of St. Louis

12/2022